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Dynamic breast response under different running intensities: a pilot finite element study across representative breast types

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ABSTRACT

This study investigates the dynamic breast response of nine representative breast types by finite element method, focusing on the effects of breast volume, running speed, and nipple spatial configuration. The simulation revealed that greater breast volume and higher speed significantly increased displacement and acceleration ($p < 0.001$). The volume and speed were key predictors of breast motion in both Y- and Z-directions, while nipple position type modulated lateral dynamics. Notably, middle-type breasts exhibited greater lateral stability. These findings offer biomechanical insights into breast movement, and provide basis for the design of sports bras tailored to different breast morphologies.

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dynamic breast response;
finite element simulation;
sports bra

1. Introduction

With increasing participation of women in running and other physical activities (Bø and Nygaard 2020), the associated breast motion has garnered significant research attention. Anatomically unsupported by skeleton and reliant on skin and ligaments (Galbreath and Goswami 2025), the breast undergoes substantial displacement during movement (Lee et al. 2022). This motion often leads to discomfort or pain by stretching connective tissues (Scurr et al. 2010, 2011; Sheng et al. 2021) and may contribute to ligament strain and tissue laxity over time (McGhee and Steele 2023), adversely affecting exercise experience. Breast dynamics are influenced by morphology, tissue properties, and movement intensity, resulting in varied biomechanical responses—especially when wearing sports bras. Age-related changes, such as increased ptosis and laxity (McGhee and Steele 2020), heighten susceptibility to deformation and discomfort in older women. Moreover, larger breast volume correlates with greater displacement and acceleration during high-impact activities like running and jumping (Ching-Sui et al. 2017; Gilmer et al. 2024), elevating risks of discomfort and tissue damage. Increased running speed further amplifies displacement amplitude and vibration frequency (Leme et al. 2020), intensifying mechanical

loading. Therefore, investigating breast dynamics during running is essential for protecting breast health, enhancing exercise comfort, and informing the design of effective supportive sportswear.

The finite element method (FEM) is widely used in biomechanics to study the dynamic motion and mechanical behavior of biological tissues. In breast kinematics, FEM simulates breast displacement, stress, and deformation during movement. Chen et al. (2013) developed a 3D FEM to simulate nonlinear breast motion during physical activity, validated by comparing simulated trajectories with motion capture data from walking, jogging, and stair stepping. Cai et al. (2018) proposed a model based on breast vibration data from free-fall experiments, and Zhang et al. (2022) developed a nonlinear 3D FEM to simulate soft tissue deformation during arm abduction. Chen et al. (2024) integrated 4D scanning and motion capture technologies to create a geometric model of the breast and surrounding tissues, using a nonlinear dynamic FEM for biomechanical analysis.

Finite element models have been developed to simulate breast–bra contact, analyzing deformation and pressure distribution under support. Sun et al. (2019) proposed a personalized FEM system to simulate bras' shaping effects and pressure distribution.

They further developed individualized FEMs based on subject geometry (Sun et al. 2021), studying how different materials and fabrics affect breast contour and pressure. Their findings showed stiffer materials reshape the breast and reduce strap pressure on the neck, though increasing pressure on the lower breast. Liu et al. (2023) introduced a novel optimization method combining CAD, FEM, 3D scanning, and boundary definition to simulate breast deformation during jumping with a sports bra. Liang et al. (2019) used FEM to investigate breast displacement, estimating damping ratios and natural frequencies for unsupported and supported breasts.

Most finite element studies on breast motion focus on specific types or movements, with limited research on diverse morphologies and running intensities. While sports bras are shown to reduce displacement and improve comfort, designs often rely on empirical experience rather than scientific evidence. Finite element modeling can provide biomechanical insights to support the creation of personalized, functional sports bras.

This study aims to employ finite element modeling technology to construct breast models for women with different breast morphologies, focusing on simulating the dynamic response characteristics of breasts during running under unconstrained conditions. By systematically analyzing the effects of different breast morphologies and running speeds on breast kinematic trajectory, relative displacement, and acceleration, the research reveals the deformation patterns and key influencing factors of breasts during unconstrained motion. This study systematically analyzes the biomechanical behavior of the breast during running under bare-chested conditions, revealing its kinematic characteristics and deformation patterns. The findings provide empirical evidence and data-driven support for the functional design, structural optimization, and improved fit of sports bras tailored to different breast morphologies and exercise intensities.

2. Materials and methods

In this study, a finite element approach was used to develop three-dimensional breast models representing nine typical morphologies, aiming to simulate their dynamic responses at running speeds between 4 and 8 km/h. To avoid intra-subject correlation, only the right breast of each participant was selected for analysis. Accordingly, the findings should be interpreted as indicative trends rather than population-level conclusions. Point cloud data of subjects with different breast types were first obtained through 3D scanning

experiments. The breast geometries were reconstructed using reverse engineering software and segmented into three components: breast tissue, soft tissue layer, and rigid torso. The breast tissue was modeled using a Mooney–Rivlin nonlinear hyperelastic constitutive material model. To balance computational efficiency and accuracy, tetrahedral elements with a mesh size of 4 mm were used for meshing. Motion-driving data were derived from motion capture experiments, in which a motion capture system recorded the kinematic parameters of running. These data were used to drive the rigid torso under gravity, enabling realistic simulation of breast dynamics. The dynamic responses of different breast types under various levels of physical intensity were analyzed systematically. An overview of the methodological workflow is illustrated in Figure 1.

2.1. Experimental methods

2.1.1. Three-dimensional body scanning

Zhang et al. (2022) proposed a breast shape classification method based on detailed morphological parameters, combining local and global breast features. Using 3D point cloud data from 158 unmarried female college students (ages 18–25), they identified eighteen parameters and selected six for cluster analysis, resulting in nine breast types with 97.33% accuracy. In this study, we applied Zhang et al.'s classification to investigate the dynamic response of different breast types during running at varying speeds. Nine representative breast types were established, integrating breast volume and nipple position, with FC, UO, and PO representing small, medium, and large volumes, and G, M, and S indicating gathered, middle, and separated nipple positions.

Nine female university students with varying breast sizes were recruited as participants for the experiment, with a mean age of 21. These individuals were also included in the initial Zhang study. Their basic anthropometric information is presented in Table 1. Participation in this study was voluntary, informed consent was obtained from all subjects, and the study protocol was approved by the Ethics Committee of Zhejiang Sci-Tech University. A 3D body scanning system ([TC]², USA) was used to capture the breast surface geometry under braless conditions, generating point cloud data for subsequent modeling. During the scanning process, uniform lighting and a stable environment were maintained to minimize shadowing and reflective interference. Participants were

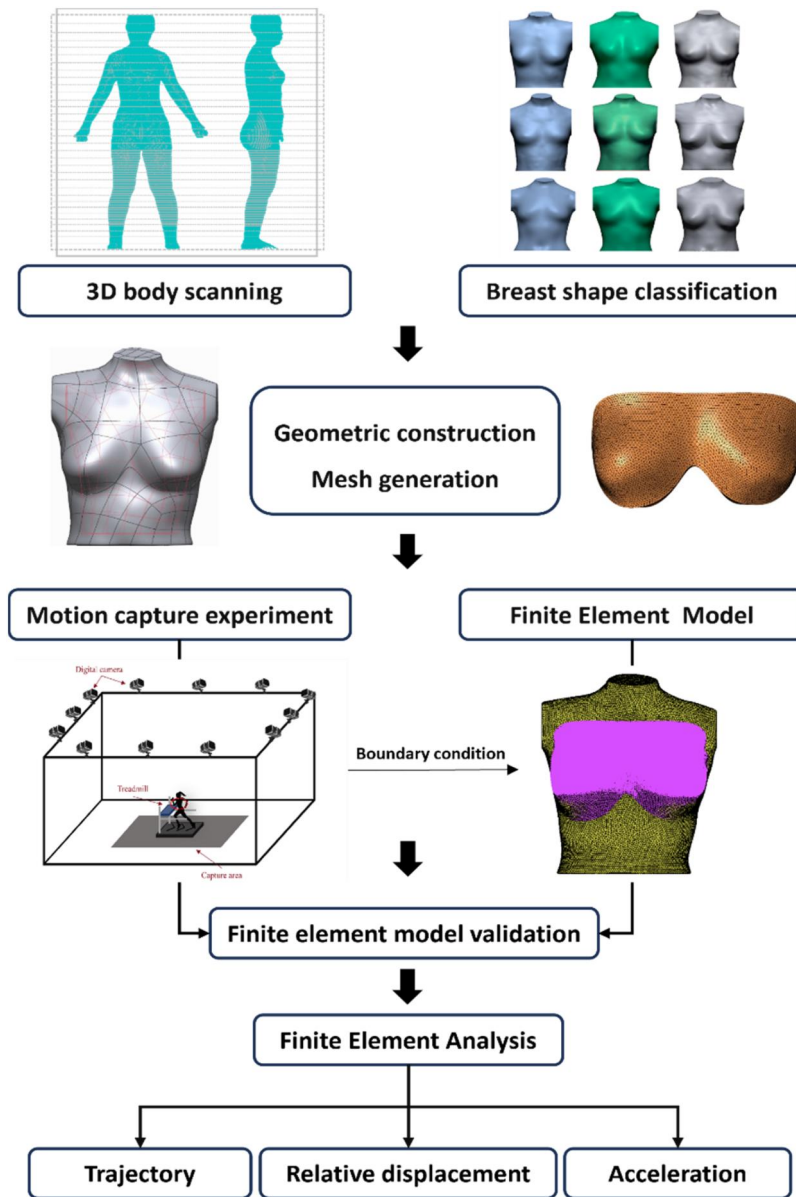


Figure 1. Schematic diagram of the overall methodological workflow.

Table 1. Basic information of participants.

ID	Height (cm)	Weight (kg)	Bust circumference (cm)	Underbust circumference (cm)	Bust difference (cm)	Cup size	Volume (ml)	Breast type
1	156.4	53.05	86.1	77.0	9.1	75A	170	FC-G
2	162.0	47.00	82.9	72.7	10.2	70B	220	UO-G
3	163.4	58.45	86.5	75.5	11.0	75B	280	PO-G
4	162.2	49.60	85.3	74.9	10.4	75B	180	FC-M
5	158.0	49.00	83.7	74.2	9.5	75A	200	UO-M
6	163.0	52.50	86.0	70.0	16.0	70D	370	PO-M
7	162.0	62.30	84.9	76.6	8.3	75A	180	FC-S
8	160.0	50.30	83.5	73.5	10.0	75B	240	UO-S
9	148.2	46.85	82.1	66.7	15.4	65C	350	PO-S

instructed to remain still and avoid muscle tension or postural changes to ensure data accuracy.

2.1.2. Three-dimensional motion capture experiment

A three-dimensional motion capture and analysis system equipped with 12 high-resolution digital cameras

was used to record breast motion across five running speeds (4 km/h, 5 km/h, 6 km/h, 7 km/h, and 8 km/h). Participants performed all trials without wearing a bra. Due to discomfort and pain, most subjects were unable to sustain running beyond 8 km/h in a braless condition. Participation in this study was voluntary, informed consent was obtained from all subjects, and the study protocol was approved by the Ethics Committee of Zhejiang Sci-Tech University. Reflective markers were placed at the suprasternal notch to track trunk motion, and at the nipples of both breasts to validate the finite element model outputs. Prior to each trial, camera calibration and coordinate system setup were conducted, and participants were instructed to remain still at the start of each motion cycle. Before the dynamic capture experiment, the treadmill was set to the target speed. Data collection began once the participant reached and maintained the target speed for approximately 15 s, with data being recorded for 1 min. The speed sequence was fixed, starting from a lower speed and gradually increasing to a higher speed. A schematic representation of the motion capture setup is shown in Figure 2.

2.2. Construction of the human finite element model

2.2.1. Geometric construction

To construct an accurate human geometric model, a multi-view optical triangulation-based 3D scanning system was employed for data acquisition. The captured point cloud was imported into Geomagic DesignX, where outliers were removed and uniform resampling was applied to dense regions to smooth the point set. A surface mesh was then generated using the 'Surface' module, with mesh density and curvature sensitivity adjusted to balance geometric accuracy and computational efficiency. Complex anatomical regions

were refined with the 'Precise Surface' tool, missing areas were reconstructed, and mesh topology was improved *via* the 'Relax Mesh' function. Finally, a deviation analysis was conducted to quantify reconstruction errors, followed by iterative correction of regions exceeding tolerance, thereby systematically enhancing the geometric fidelity of the model. The upper torso was simplified into three main components: deformable breasts, a soft tissue layer, and a rigid torso. As illustrated in Figure 3(b), breast geometries were extracted through segmentation procedures. Following breast removal, the skeletal structure was generated *via* surface offsetting (set to 23 mm to represent the soft tissue thickness), resulting in the geometry shown in Figure 3(a). An automatic surface generation process was then applied to derive the soft tissue model, as shown in Figure 3(c). Boolean operations were applied to generate solid models of the breast, soft tissue, and skeletal regions. The torso was defined as a non-deformable rigid body to which dynamic displacement boundary conditions were applied to drive model motion, while the breast and soft tissue were modeled as deformable three-dimensional structures. A compensation-based iterative algorithm [16] was applied to remove the gravitational deformation induced during 3D scanning, thereby restoring gravity-free breast geometry for accurate simulation.

2.2.2. Mesh generation

In finite element modeling, breast geometry was discretized using a tetrahedral mesh to simulate mechanical behavior. Mesh density affects both simulation accuracy and computational cost, requiring a balance between precision and efficiency (Liang et al. 2019). Due to the anatomical complexity of the breast, hexahedral meshing is often impractical (McCurry et al. 2015), making tetrahedral elements a common choice in previous studies (del Palomar et al. 2008; Han

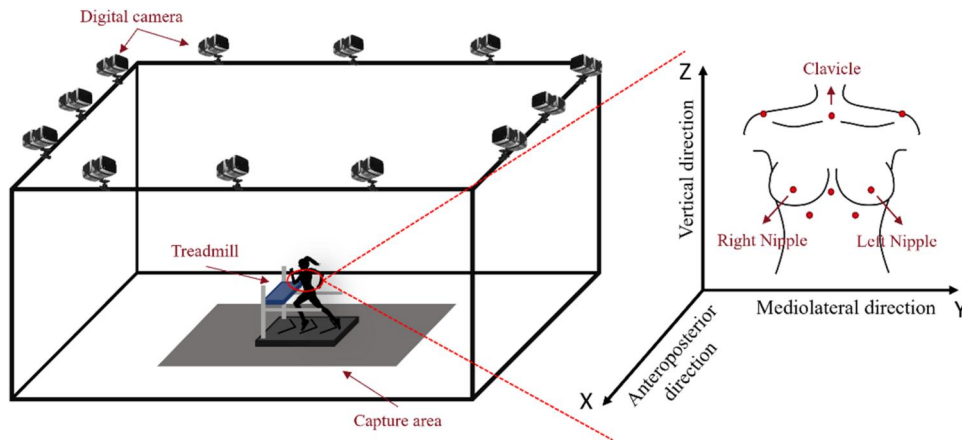


Figure 2. Schematic diagram of the 3D motion capture experiment.

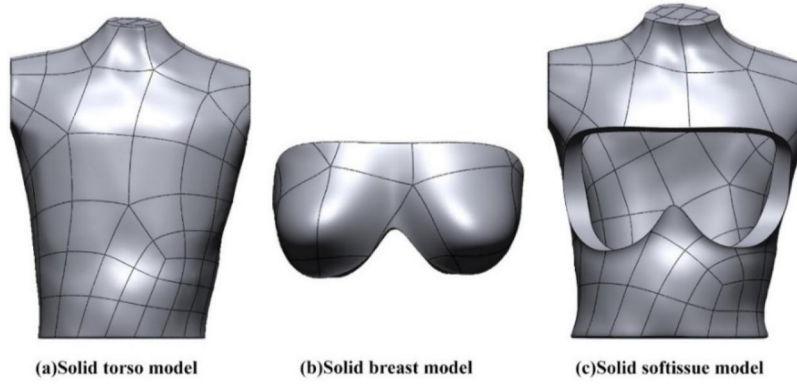


Figure 3. Geometric model of the human body.

et al. 2010; Chen et al. 2013; Sun et al. 2019). Typical element sizes in the literature range from 4 mm (Roose et al. 2005) and 5 mm (Sun et al. 2019, 2021) to 6 mm (Liang et al. 2019) and 8 mm (Liu et al. 2023). To balance computational efficiency and numerical accuracy, a tetrahedral element size of 4 mm was selected in this study. Both the soft tissue and breast components were discretized into finite tetrahedral elements, with node-sharing implemented at their interface to prevent separation during motion.

2.2.3. Material definition

In this study, identical material properties (hardness, stiffness, and other mechanical properties) were assigned to all breast models. This assumption was made to standardize variables and focus on the effects of breast morphology and running intensity on dynamic breast response characteristics. The breast tissue was modeled as a hyper-elastic material capable of undergoing large deformations. Among the hyper-elastic models commonly used to characterize breast material properties are the Mooney–Rivlin model (Pianigiani et al. 2015; Liang et al. 2019; Sun et al. 2019a,b, 2021; Zhang et al. 2022) and the Neo-Hookean model (del Palomar et al. 2008; Gamage et al. 2012; Mira et al. 2018; Chen et al. 2019).

The strain energy density function W can be expressed as,

$$W(I_1, I_2) = \sum_{i,j=0}^n C_{ij} (I_1 - 3)^i (I_2 - 3)^j$$

where C_{ij} are the hyper-elastic parameters describing the nonlinear elastic behavior of the breast tissue, and I_1 and I_2 are the first and second invariants of the left Cauchy–Green deformation tensor, respectively. The parameter n denotes the order of the material model.

In this study, the breast was assumed to follow a second-order Mooney–Rivlin model ($n=2$), which is characterized by five coefficients: C_{10} , C_{01} , C_{11} , C_{20} and C_{02} . The soft tissue was modeled as a

Table 2. Material coefficients for breast and soft tissue layers.

Component	Material model	Material parameters
Soft tissue	Nonlinear isotropic elastic material	$E = 100 \text{ kPa}$, $\nu = 0.3$
Breast	Hyper-elastic Mooney–Rivlin material	$C_{10} = 0.3 \text{ kPa}$ $C_{01} = 0.31 \text{ kPa}$ $C_{11} = 2.5 \text{ kPa}$ $C_{20} = 4.7 \text{ kPa}$ $C_{02} = 3.8 \text{ kPa}$

homogeneous, isotropic material with nonlinear behavior. The material coefficients for both the breast and soft tissue layers are listed in Table 2.

2.2.4. Boundary conditions

In defining the contact relationship, the interface between the breast soft tissue and the torso was assumed to have no relative motion; therefore, a tied contact was applied between the breast and torso models. Regarding boundary conditions, the breast model was initially placed in a gravity-free state. Gravitational loading was then applied to simulate the realistic mechanical response of the breasts during motion. On this basis, dynamic torso displacement captured *via* a three-dimensional motion tracking system was applied as a displacement boundary condition to drive the overall movement of the finite element model, as illustrated in Figure 4. This procedure ensures realistic breast–torso interaction and improves the accuracy of boundary condition representation in dynamic simulations. The experiment was conducted on a treadmill to simulate in-place running, during which displacement in the X-direction (anterior–posterior) remains nearly constant. In real-world conditions, however, X-direction displacement gradually increases over time. As breast pain is primarily associated with acceleration and velocity caused by lateral sway and vertical oscillation, this study focuses on breast deformation in the Y- and Z-directions.

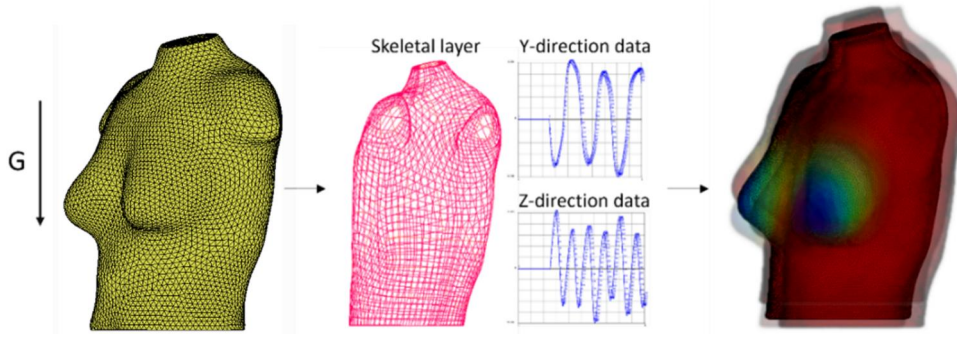


Figure 4. Schematic illustration of the applied boundary conditions.

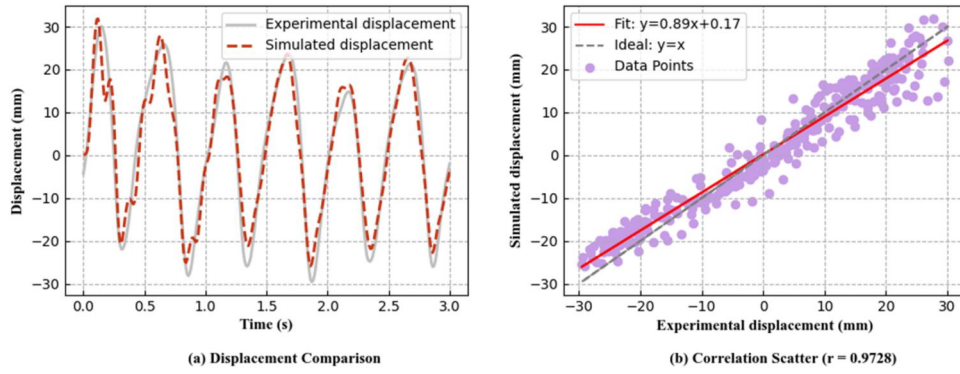


Figure 5. Experimental vs. simulated displacement: time-domain comparison and scatter plot.

3. Results and analysis

3.1. Validation of finite element model accuracy

To validate the accuracy and effectiveness of the developed finite element model, the displacement of a marker point on the right breast along the Z axis (vertical direction) was selected as the evaluation metric. Experimental measurements were compared with simulation results over a total duration of 3 s, with a sampling interval of 0.01 s, yielding 300 frames of continuous displacement data.

To quantify the discrepancy between the two datasets, both absolute and relative errors were evaluated using the root mean square error (RMSE) and relative root mean square error (RRMSE), respectively. The corresponding formulas are presented in Equations (1) and (2). Additionally, the Pearson Correlation Coefficient was employed to assess the consistency of the temporal trends between the two datasets, as shown in Equation (3).

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (D_{EXPi} - D_{FEMi})^2} \quad (1)$$

$$RRMSE = \frac{RMSE}{\text{MAX}(D_{EXP}) - \text{MIN}(D_{EXP})} \times 100\% \quad (2)$$

$$r = \frac{\sum_{i=1}^N (D_{EXPi} - \overline{D_{EXPi}})(D_{FEMi} - \overline{D_{FEMi}})}{\sqrt{\sum_{i=1}^N (D_{EXPi} - \overline{D_{EXPi}})^2 \sum_{i=1}^N (D_{FEMi} - \overline{D_{FEMi}})^2}} \quad (3)$$

Here, D_{EXPi} and D_{FEMi} denote the experimental measurement and the finite element simulation result at time frame i , respectively. $\overline{D_{EXPi}}$ and $\overline{D_{FEMi}}$ represent their respective mean values. N is the total number of sampling points. $\text{MAX}(D_{EXP})$ and $\text{MIN}(D_{EXP})$ refer to the maximum and minimum values in the sequence of experimental measurements, respectively.

Figure 5(a) compares the simulated and experimentally measured breast displacements over time. The close match in trend, oscillation, and peak positions between the experimental data and simulation indicates that the finite element model reliably captures the temporal dynamics of breast motion. Figure 5(b) presents a scatter plot of simulated versus measured values, with points distributed along the diagonal, demonstrating strong linear agreement. This visual correlation supports the statistical validation of the model and confirms its predictive accuracy.

Further analysis revealed that the root mean square error (RMSE) between the simulated and experimental displacements was 4.01 mm, while the relative root mean square error (RRMSE), calculated according to

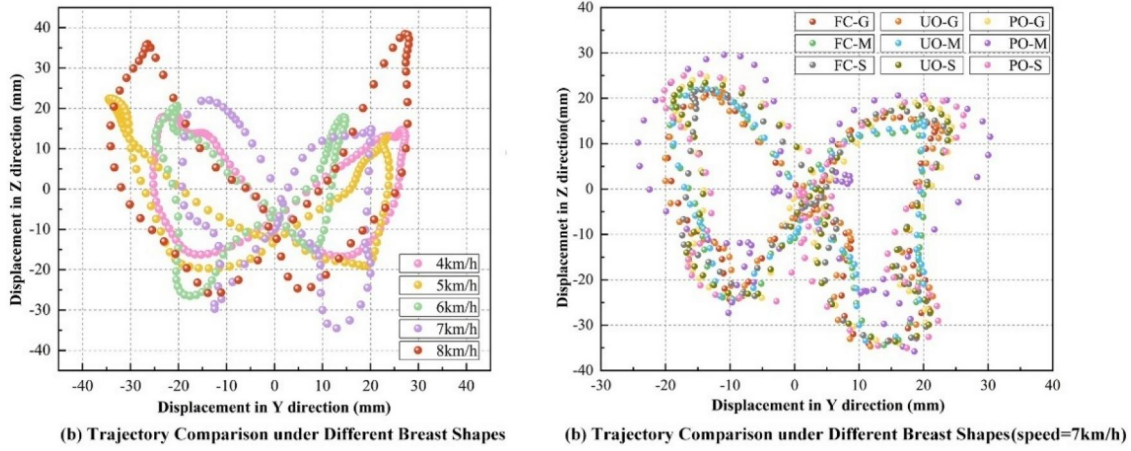


Figure 6. Comparison of nipple motion trajectories across different running speeds and breast types.

Equation (2), was 6.72%. In addition, the Pearson correlation coefficient between the simulation and experimental datasets reached 0.973, indicating a high degree of consistency and strong linear correlation in both temporal trends and numerical distributions.

3.2. Comparative analysis of breast motion trajectories under different breast types and running speeds

Finite element simulation results indicate that during running, the trajectory of the nipple exhibits an irregular lateral '∞' shaped pattern, as illustrated in Figure 6. Figure 6(a) shows the motion trajectories of the UO-M breast type under different running speeds, while Figure 6(b) presents the trajectories of various breast types at a running speed of 7 km/h.

Simulation results show that as running speed increases from 4 to 8 km/h, breast displacement grows from 47.92 to 82.09 mm laterally (Y) and from 44.02 to 99.17 mm vertically (Z). The motion pattern also shifts: lateral displacement dominates at lower speeds ($\sim 1.55 \times$ vertical at 4–5 km/h), while at higher speeds (6–8 km/h) the two directions become nearly equal (ratio ~ 0.9). Breast volume strongly affects displacement magnitude, whereas nipple position has little influence during running.

3.3. Regression analysis of maximum relative displacement under different conditions

The relative displacement of the nipple point (D_r) is defined as its displacement relative to the torso, calculated by subtracting the displacement of the clavicle point from that of the nipple point, as expressed in

Equation (4).

$$\vec{D}_r = \vec{D}_{nipple} - \vec{D}_{clavicle} \quad (4)$$

Here, D_{nipple} and $D_{clavicle}$ represent the displacements of the nipple point and the clavicle point, respectively.

Figure 7 illustrates the relative displacement (D_r) of the nipple point for nine different breast types obtained from finite-element simulations. A comparison of the lateral (Y axis) and vertical (Z axis) relative displacements across running speeds of 4–8 km/h under bare-chested conditions reveals that the largest breast volume reached a maximum lateral displacement of 20.77 mm at 7 km/h, while its maximum vertical displacement was 37.12 mm at 8 km/h. Overall, breast motion exhibits a clear speed-dependent transition: at low-to-medium speeds (4–6 km/h), lateral displacement generally exceeds vertical displacement; however, when speed increases to 7–8 km/h, vertical displacement surpasses lateral displacement, indicating that vertical oscillation becomes more pronounced with increasing speed. This shift likely reflects nonlinear dynamic behavior arising from enhanced inertial effects, the nonlinear tissue mechanics described by the Mooney–Rivlin model, and gait-induced loading changes at higher speeds.

A multiple linear regression analysis was conducted to assess the influence of breast volume, running speed, and nipple position type (categorized as Gathered, Middle, or Separated) on maximum relative nipple displacement in the Y- and Z-directions. Position type was dummy-coded with Gathered as the reference; coefficients are reported only for Middle and Separated. As shown in Figures 8(a) and 8(b)—where data points are colored by position type—

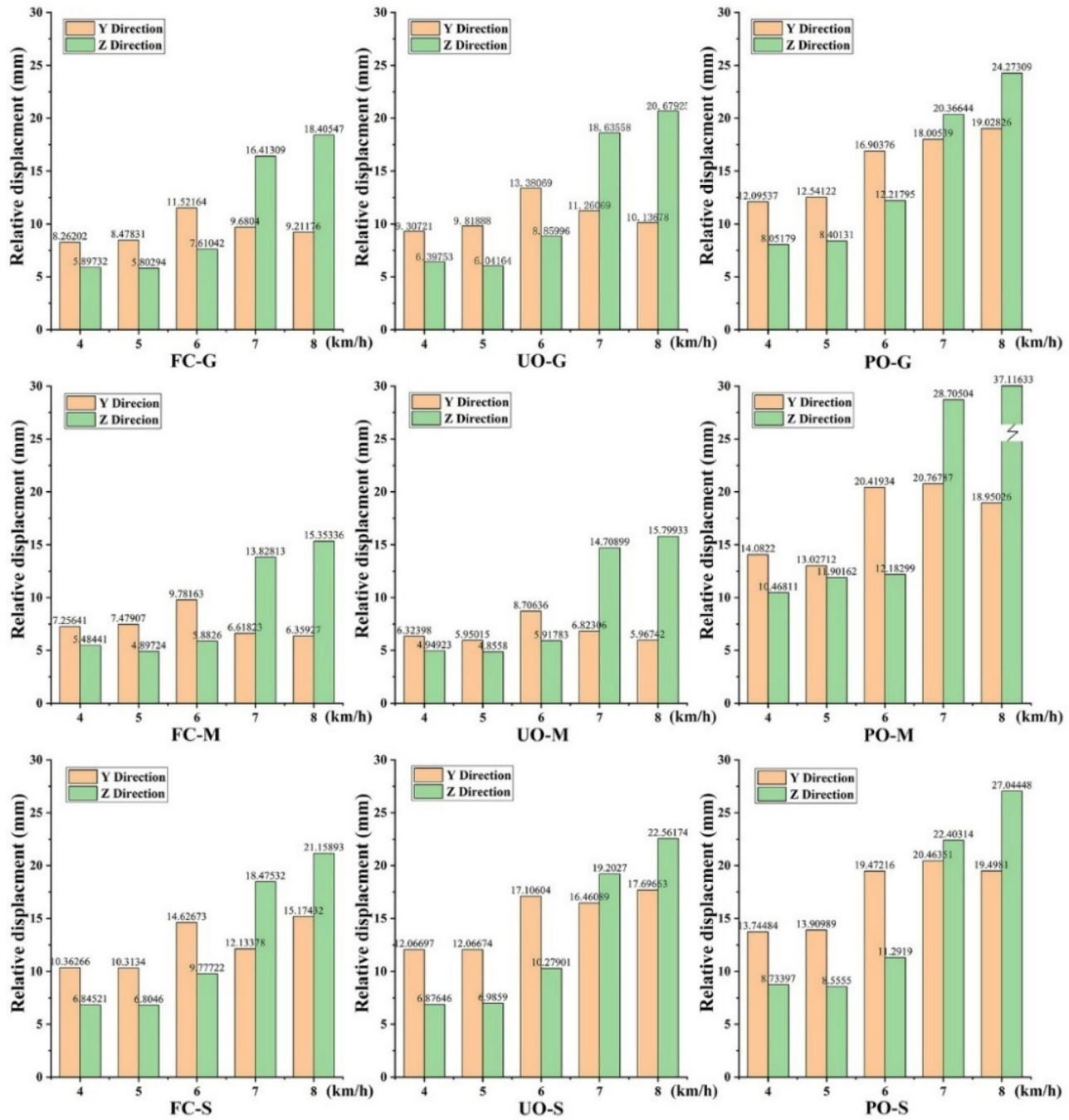


Figure 7. Comparison of nipple relative displacement across different breast types and running speeds.

displacement increases with breast volume in both directions, with a stronger effect in the Y-direction. Full regression results are provided in Tables 3 (Y-direction) and 4 (Z-direction).

Figure 8(a) illustrates the relationship between breast volume and maximum relative displacement in the Y-direction across nipple-position types (Gathered, Middle, Separated). All three types show increasing displacement with volume, with the Separated type yielding the highest values and the Middle type the lowest. Multiple linear regression

(Table 3; $R^2 = 0.784$) confirms significant positive effects of volume (coefficient = 0.0462, $p < 0.001$) and speed (coefficient = 0.9519, $p < 0.001$). Compared to the Gathered type, the Middle type shows significantly lower displacement (coefficient = -2.7177 , $p = 0.002$), whereas the Separated type does not differ significantly ($p = 0.077$), suggesting that the Middle type provides better lateral stability, likely due to its more centralized nipple distribution.

Figure 8(b) shows a similar volume-displacement trend in the Z-directions, though differences among

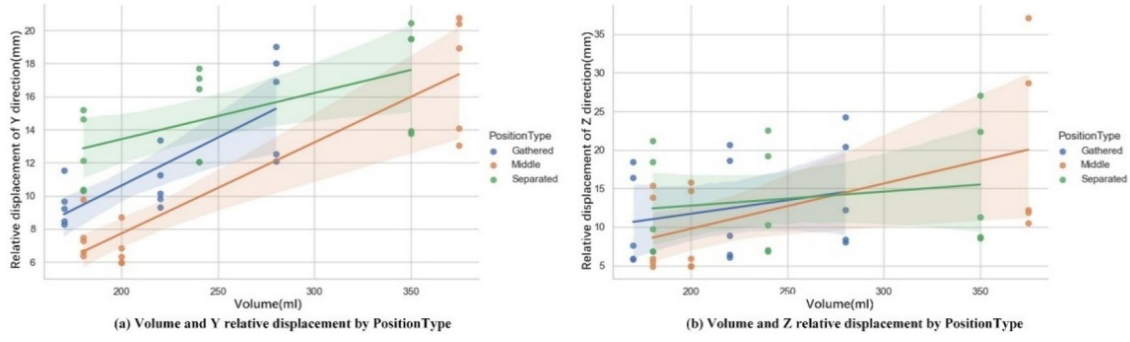


Figure 8. Regression trends between breast volume and maximum relative displacement in the Y- and Z-directions across different nipple spatial position types.

Table 3. Multiple linear regression results for maximum relative displacement in the Y-direction ($R^2 = 0.784$).

Variables	Coefficient	Standard error	<i>t</i> value	<i>p</i> value
Intercept	−4.059	1.836	−2.211	0.033
C (PositionType-M)	−2.718	0.816	−3.331	0.002**
C (PositionType-S)	1.490	0.820	1.817	0.077*
Volume	0.462	0.005	9.831	<0.001***
Speed	0.9519	0.232	4.097	<0.001***

Note. *PositionType* is a categorical variable with *Gathered* as the reference group. The reported coefficients represent estimates relative to this baseline category.

Table 4. Multiple linear regression results for maximum relative displacement in the Z-directions ($R^2 = 0.810$).

Variables	Coefficient	Standard error	<i>t</i> value	<i>p</i> value
Intercept	−22.4858	2.883	−7.800	<0.001
C (PositionType-M)	−0.9131	1.281	−0.713	0.480
C (PositionType-S)	−0.1249	1.288	−0.097	0.923
Volume	0.0416	0.007	5.638	<0.001***
Speed	4.2874	0.365	11.750	<0.001***

Note. *PositionType* is a categorical variable with *Gathered* as the reference group. The reported coefficients represent estimates relative to this baseline category.

position types are less marked. Regression results (Table 4; $R^2 = 0.810$) indicate strong contributions from volume (coefficient = 0.0416, $p < 0.001$) and speed (coefficient = 4.2874, $p < 0.001$). In contrast to the Y-direction, neither the Middle ($p = 0.480$) nor the Separated ($p = 0.923$) type differs significantly from the Gathered reference, indicating that vertical motion is driven primarily by volume and movement intensity rather than nipple spatial configuration.

In summary, breast volume and running speed are the primary determinants of nipple displacement in both directions, with speed being especially dominant vertically. Nipple position significantly modulates lateral (Y-direction) displacement—the Middle type exhibits superior stability—but has negligible effect on vertical displacement.

These findings provide theoretical support for the structural optimization of supportive garments, such as sports bras, emphasizing the significant modulating role of nipple spatial configuration in suppressing

lateral breast motion. Personalized bra designs should be tailored to the dynamic characteristics of different breast types and the intensity of physical activity, enhancing stability and comfort, particularly during high-speed and high-impact activities.

3.4. Regression analysis of maximum acceleration under different conditions

In this study, acceleration of the nipple point was calculated based on displacement data obtained from finite element simulations using a numerical differentiation method. Specifically, velocity and acceleration were derived by computing the second-order time derivative of displacement. The resulting acceleration values, originally expressed in mm/s^2 , were converted to m/s^2 and subsequently normalized by gravitational acceleration (g) for comparative purposes. The formula for acceleration calculation is provided in Equation (5).

$$a(t) = \frac{d^2D(t)}{dt^2} \quad (5)$$

Here, $D(t)$ represents the displacement as a function of time, and $a(t)$ denotes the instantaneous acceleration of the nipple point.

To reduce high-frequency noise introduced by numerical differentiation, a zero-phase fourth-order Butterworth low-pass filter with a cutoff frequency of 6 Hz was applied to the acceleration signals for smoothing, thereby enhancing the stability and accuracy of the results.

Figure 9 shows the maximum nipple acceleration for nine breast types under bare-chested conditions across speeds of 4–8 km/h. At moderate-low speeds (4–6 km/h), certain types (e.g. FC-M, UO-G) exhibited higher lateral (Y axis) acceleration, indicating dominant horizontal oscillation. The largest-volume breast type reached the highest vertical (Z axis)

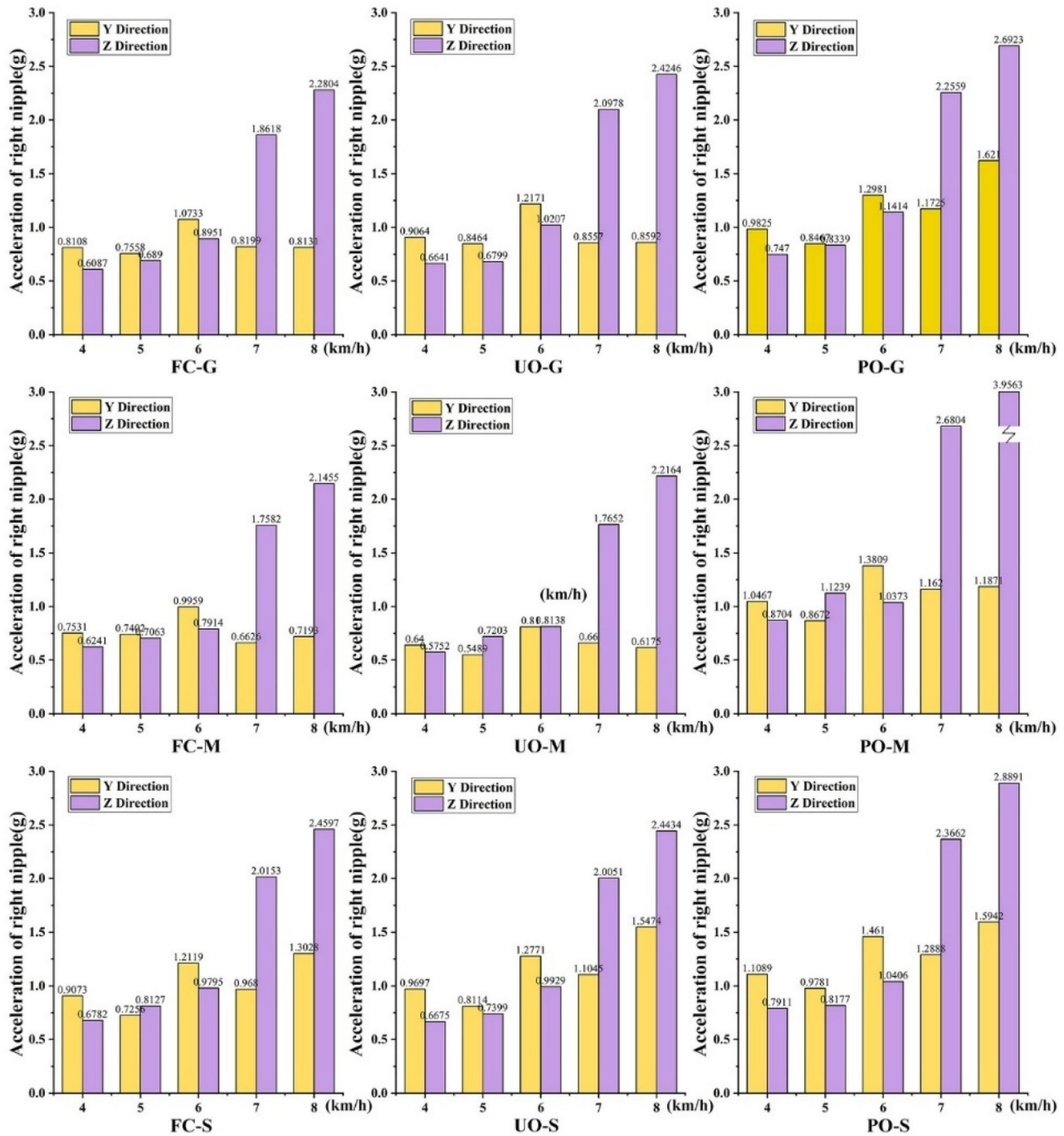


Figure 9. Comparison of nipple acceleration across different breast types and running speeds.

acceleration of 3.9563 g at 8 km/h, reflecting intense vertical vibration.

Trends in acceleration direction were consistent: at 4–6 km/h, several types showed slightly higher Y- than Z-directions acceleration, while at 7–8 km/h, Z-directions acceleration increased markedly and exceeded Y-direction values in most types. This shift suggests that vertical oscillation becomes the primary risk factor as running speed intensifies.

Multiple linear regression was conducted to analyze the effects of breast volume, running speed, and

nipple spatial position (PositionType: Gathered, Middle, Separated) on the maximum nipple acceleration in the Y- and Z-directions. PositionType was dummy-coded with Gathered as the reference; regression tables therefore report coefficients only for Middle and Separated. Each model reports the coefficient, standard error, *t* statistic, *p* value, and the coefficient of determination (R^2) to assess predictor significance and overall fit.

Scatter regression plots (Figures 10(a) and 10(b)) visually illustrate the relationship between breast

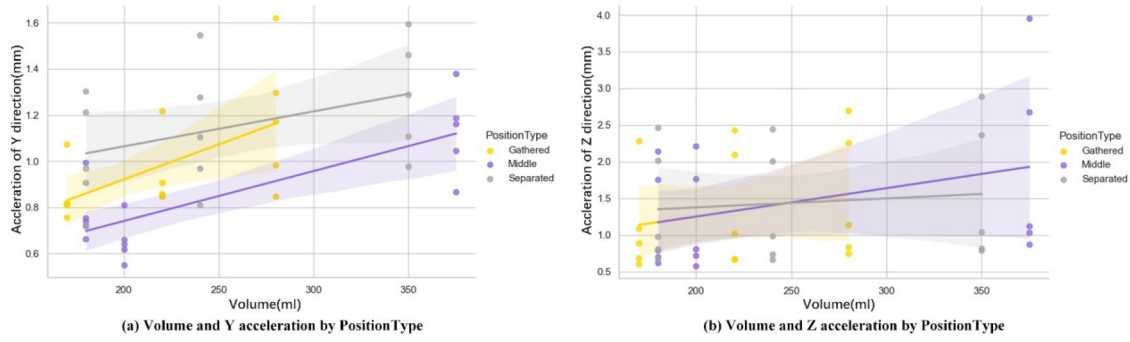


Figure 10. Regression trends between breast volume and maximum acceleration in the Y- and Z-directions across different nipple spatial position types.

Table 5. Multiple linear regression results for maximum acceleration in the Y direction ($R^2 = 0.601$).

Variables	Coefficient	Standard error	t value	p value
Intercept	0.1405	0.151	0.930	0.358
C (PositionType-M)	-0.1977	0.067	-2.944	0.005**
C (PositionType-S)	0.0896	0.068	1.328	0.192
Volume	0.0021	0.000	5.340	<0.001***
Speed	0.0650	0.019	3.396	0.002**

Note. PositionType is a categorical variable with Gathered as the reference group. The reported coefficients represent estimates relative to this baseline category.

Table 6. Multiple linear regression results for maximum acceleration in the Z-directions ($R^2 = 0.823$).

Variables	Coefficient	Standard error	t value	p value
Intercept	-2.3526	0.306	-7.692	0.000
C(PositionType-M)	0.0266	0.136	0.196	0.846
C(PositionType-S)	0.0061	0.137	0.044	0.965
Volume	0.0030	0.001	3.791	<0.001***
Speed	0.5051	0.039	13.049	<0.001***

Note. PositionType is a categorical variable with Gathered as the reference group. The reported coefficients represent estimates relative to this baseline category.

volume and maximum acceleration in the Y and Z directions, with points colored by PositionType and fitted lines highlighting type-specific trends. Detailed numerical results are provided in Tables 5 (Y-direction) and 6 (Z-directions), which list all predictors' coefficients, standard errors, t values, and p values for comparing the significance and direction of each factor's influence.

Figure 10(a) illustrates the relationship between breast volume and maximum Y-direction acceleration across nipple-position types (Gathered, Middle, Separated). Acceleration increases with volume in all types, with the Separated type showing higher baseline values at larger volumes. Multiple linear regression (Table 5; $R^2 = 0.601$) confirms significant positive effects of volume (coefficient = 0.0021, $p < 0.001$) and speed (coefficient = 0.0650, $p = 0.002$). Compared to Gathered, the Middle type exhibits significantly lower acceleration (coefficient = -0.1977, $p = 0.005$), while the

Separated type shows a non-significant increase ($p = 0.192$). This indicates that nipple position modulates lateral acceleration, with the Middle type providing better suppression, likely due to its centralized distribution and more uniform load transfer.

Figure 10(b) shows a similar volume-acceleration trend in the Z-directions, though differences among position types are small. The regression model fits well ($R^2 = 0.823$; Table 6), with volume (coefficient = 0.0030, $p < 0.001$) and speed (coefficient = 0.5051, $p < 0.001$) being strong predictors. PositionType has no significant effect on vertical acceleration ($p > 0.8$), indicating that nipple spatial configuration plays a minor role in vertical dynamics.

In summary, breast volume and running speed are the primary drivers of nipple acceleration in both directions, with speed being especially dominant vertically. Nipple position significantly influences lateral (Y-direction) acceleration—the Middle type shows superior stability—but has negligible effect on vertical motion. These findings offer theoretical support for optimizing support garments, such as sports bras, particularly for controlling lateral breast movement during high-impact activities.

5. Discussion

This study investigates the dynamic breast response during running across different breast types using finite element modeling. The results underscore that both breast volume and running speed are significant determinants of breast displacement and acceleration. Running speed, particularly in the vertical (Z-directions), has a particularly dominant effect, with greater displacement observed at higher speeds. Nipple spatial position plays a crucial role in modulating lateral (Y-direction) displacement, with the 'Middle' nipple position providing greater stability, likely due to its more compact anatomical structure.

For a given breast volume, breast displacement and acceleration increased with running speed within the investigated range, indicating a strong dependence on motion intensity. At higher speeds, slight deviations from linearity were observed, suggesting that the dynamic response of breast tissue becomes more complex under increased inertial loading. The Mooney–Rivlin material model, which was used to describe nonlinear tissue properties, may contribute to this nonlinear behavior. Future studies could further explore the detailed mechanisms behind these nonlinear effects and their impact on breast dynamics under different motion conditions.

However, several limitations must be considered when interpreting these findings. First, the small sample size ($n=9$) and the focus on a specific demographic—young, college-aged women (18–25)—limit the generalizability of the results. This sample may not fully represent the dynamic breast behaviors of older women, whose breast tissue is more likely to experience ptosis and laxity, potentially resulting in distinct displacement patterns. Moreover, the study focused solely on the right breast and did not account for asymmetry between the left and right breasts, which could influence overall displacement and affect the accuracy of the results. The assumption of uniform material properties across all participants also overlooks individual differences in breast tissue stiffness and elasticity, which may significantly impact breast motion during high-speed activities.

Given these limitations, future research should expand the sample size and include a more diverse population, spanning a broader age range, to better understand how breast dynamics change with age. Incorporating both breasts in the analysis to account for asymmetry would also provide a more comprehensive view of breast motion. Additionally, refining the model to incorporate subject-specific material properties would enhance the accuracy of simulations, particularly in light of the variability in breast tissue across individuals. Finally, extending the research to explore non-linear dynamic behaviors at higher speeds and in more complex movements, such as jumping or cutting, would offer valuable insights into optimizing sports bras for different activities, providing better support and comfort for a broader range of users.

6. Conclusion

This study systematically explored the dynamic breast response characteristics of nine representative female

breast types under varying running speeds, through the integration of finite element modeling and multiple linear regression analysis. The simulation results demonstrated strong consistency with experimental measurements, validating the model's accuracy and stability in predicting breast dynamics during locomotion. The results revealed that breast volume and running speed are key factors influencing maximum displacement and acceleration, both showing significant positive correlations in the Y- and Z-directions. Notably, speed exerted a stronger effect on vertical (Z-directions) displacement, highlighting the critical role of impact loading in high-intensity movement. Nipple spatial position type exhibited a moderating effect on Y-direction responses, with the *Middle* type showing superior lateral stability in terms of both displacement and acceleration. In contrast, differences among position types were not significant in the Z-directions, suggesting that vertical vibration is primarily driven by physical properties rather than anatomical configuration. Collectively, this integrated simulation and statistical analysis framework not only advances the understanding of female breast biomechanics during movement but also provides a scientific basis for the structural optimization, parameter regulation, and fit customization of sports bras—offering theoretical support and a technical pathway for the design of personalized wearable products.

Authors contributions

CRedit: **Jie Yu**: Investigation, Visualization, Writing – original draft; **Yue Sun**: Conceptualization, Funding acquisition, Methodology, Supervision; **Shichen Zhang**: Writing – review & editing; **Kit-lun Yick**: Resources, Software; **Bingfei Gu**: Data curation, Funding acquisition, Project administration.

Disclosure statement

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